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อบรมวิชาการระยะสั้น ประจำปี 2569
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การใช้ยาปฏิชีวนะเกินความจำเป็น ในภาวะติดเชื้อในกระแสเลือดระยะแรกหลังเกิด

Antibiotic Overuse in Neonatal Early-Onset Sepsis

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Outline

- 1 Epidemiology and the treatment gap
- 2 The seven common pitfalls
- 3 Short-term and long-term consequences
- 4 Tools and strategies to reduce overuse
- 5 Summary and application to practice



Early onset sepsis (EOS): definition and importance

72

first hours of life

Why the urgency

It progresses rapidly and can cause death or permanent disability if untreated

From pre-emptive treatment to a new problem

- Past: better safe than sorry

Start antibiotics as soon as there is any suspicion, even when the probability of infection is low, for fear of a catastrophic outcome

- Present: the incidence has fallen sharply

Universal maternal GBS screening and IAP have clearly reduced EOS in term infants

- But antibiotic use has not fallen in proportion

Neonatal antibiotic use has stayed the same → overuse has become a major problem



Epidemiology: the gap between treatment and actual disease

0.2-1

cases per 1,000 live births

EOS incidence in term infants

5-10%

of all newborns

receive antibiotics within 48 hours

1-3%

of infants given antibiotics

have culture-confirmed infection

For every 100 infants treated for suspected EOS

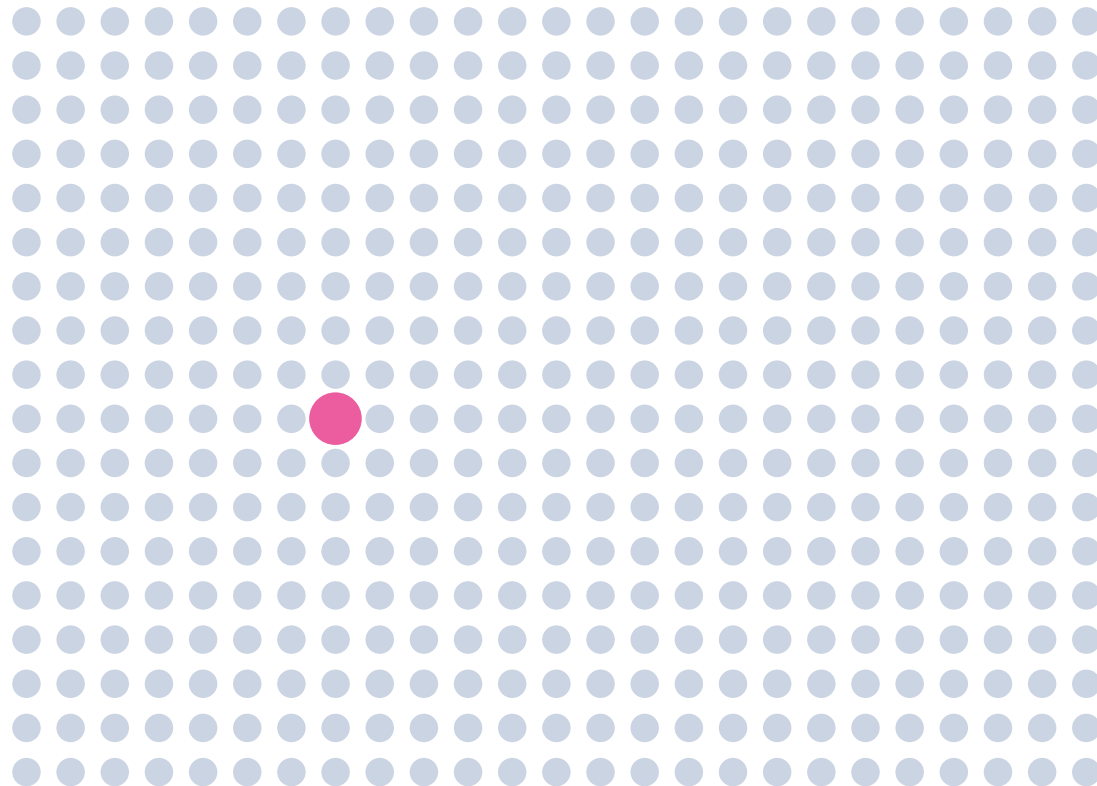
97–99 infants have no true infection

1–3 infants only have a positive culture — the rest carry the risks of antibiotics with no benefit



A mathematical imbalance: $NNT = 450$

Infants > 35 weeks' gestation whose mothers had chorioamnionitis



450

infants must receive antibiotics
to treat proven EOS in **just 1 infant**

The risk accepted in exchange

- Separation from the mother ·
- Unnecessary procedures ·
- Microbiome disruption



The seven pitfalls that lead to antibiotic overuse

The problem is not a lack of guidelines, but the way data are interpreted and clinical decisions are made

1

Maternal risk factors

≠ neonatal infection

2

Non-specific signs

after birth

3

Blood cultures

below standard

4

Biomarkers used to

start, not to stop

5

No clear criteria

for stopping

6

Fear of missing

the diagnosis

7

System-level pitfalls

and unit culture

All seven recur across settings, and many low-risk infants receive antibiotics unnecessarily



Pitfall 1: maternal risk factors $\neq$ neonatal infection

Starting antibiotics on risk factors alone and continuing 7–10 days without assessing the infant

Commonly used intrapartum risk factors

- Chorioamnionitis / intra-amniotic infection
- Maternal intrapartum fever
- Prolonged rupture of membranes (PROM) > 18 h
- Maternal GBS colonization
- Gestational age below 37 weeks

These data matter for risk assessment,
but they are not a diagnosis

Relative risk

2-8

fold (odds ratio)

Looks like high risk

Absolute risk

< 1%

in most settings

The real risk stays low

because the incidence is so low

Even though risk factors raise relative risk 2–8 fold, in term and > 35-week infants the absolute risk stays below 1%

Use them to “**assess risk**”,
not to “decide there is infection”



Pitfall 2: non-specific signs during postnatal transition

The early signs of EOS overlap with findings that are common and normal in the first 24 hours of life

? Signs that can be normal

- Transient tachypnea
- Hypoglycaemia
- Temperature instability
- Minor spitting up

! What points to true EOS

- Severe from the outset
- A clearly worsening course
- No response to basic care
- A trend moving in a negative direction

What to assess: Severity · Trend · Response to basic care

Most infants given antibiotics for non-specific signs turn out to have no true infection



Pitfall 3: substandard blood cultures

Blood culture is the gold standard — but the quality of collection determines how far the result can be trusted

≥ 1 ml

per culture set

Common errors in sample collection

- ✗ Too little blood drawn
- ✗ Blood drawn after antibiotics have started
- ✗ Skin contamination during the draw

Most EOS pathogens turn positive within 24–36 hours



Continuing antibiotics past this window when the culture is negative

usually reflects clinician uncertainty rather than clinical need



Pitfall 4: using biomarkers to start antibiotics rather than to stop them

CRP and procalcitonin have limited sensitivity and specificity in the early hours after birth

✗ Wrong use — “On-switch”

- One high CRP used to start or prolong
- Forgetting that CRP can rise from the stress of birth or from hypoxia
- Result: stopping is delayed, use rises

✓ Appropriate use — “Off-switch”

- Two consecutive negative CRPs at least 24 h apart
- Repeated negatives make EOS far less likely
- Use it to “build confidence to stop”, not to start

A serial CRP approach

0 h

Blood culture before antibiotics

12-24 h

First CRP; if negative → the risk falls

36-48 h

Second negative CRP → stop



Pitfalls 5–7: stopping criteria · fear · the system

These last three are why overuse persists even though the guidelines are already clear

5

No clear criteria for stopping

- Guidance on “starting” is clear; guidance on “stopping” is not
- AAP: stopping within 36–48 h in a well infant with a negative culture is safe practice

6

Fear of missing the diagnosis

- Fear of a catastrophic outcome is a major driver of overuse
- The harm from overuse is not obvious in the short term

7

System-level pitfalls

- A “better safe than sorry” culture of care
- No multidisciplinary antibiotic stewardship team



The consequences of antibiotic overuse

Real harm, even though it is not obvious while the infant is still in the unit

Short term

- Longer hospital stay
- More procedures
- Separation from the mother
- Higher risk of fungal infection
- Higher risk of necrotising enterocolitis (NEC)

Long term

- Disrupted immune development
- Higher risk of allergic disease
- Higher risk of asthma
- Long-term antimicrobial resistance
- Effects on the gut microbiome

These effects are rarely visible while the infant is in the unit **but they matter for long-term health**

Term infants vs very preterm infants

NNTs differ enormously — starting antibiotics in preterm infants is often right; the problem is not stopping them

Infant group	NNT
GA > 35 weeks with maternal chorioamnionitis	450
GA < 34 6/7 weeks (overall)	6-40
Low-risk preterm: C/S without labour, no ROM, no chorioamnionitis	~200
The above + no histologic chorioamnionitis	~380

The key point in very preterm infants

Starting antibiotics empirically is often appropriate; the common problem is **failing to stop them when nothing supports infection**



Tools and strategies in infants > 35 weeks' gestation

Three main approaches to the decision — differing in how finely they assess risk

1 Categorical risk assessment

- Presence/absence of risk factors
- In use for more than 20 years
- Assigns 5–10% to treatment

Limitation: treats risk factors as binary; ignores absolute risk and the evolving clinical picture

2 Multivariate (EOS risk calculator)

- Expresses risk as a numerical value
- Reduces cultures and antibiotics

Strength: moves from a binary decision to individualised, quantitative risk assessment

3 Serial clinical observation

- Systematic, repeated assessment
- EOS signs usually appear at 6–36 h

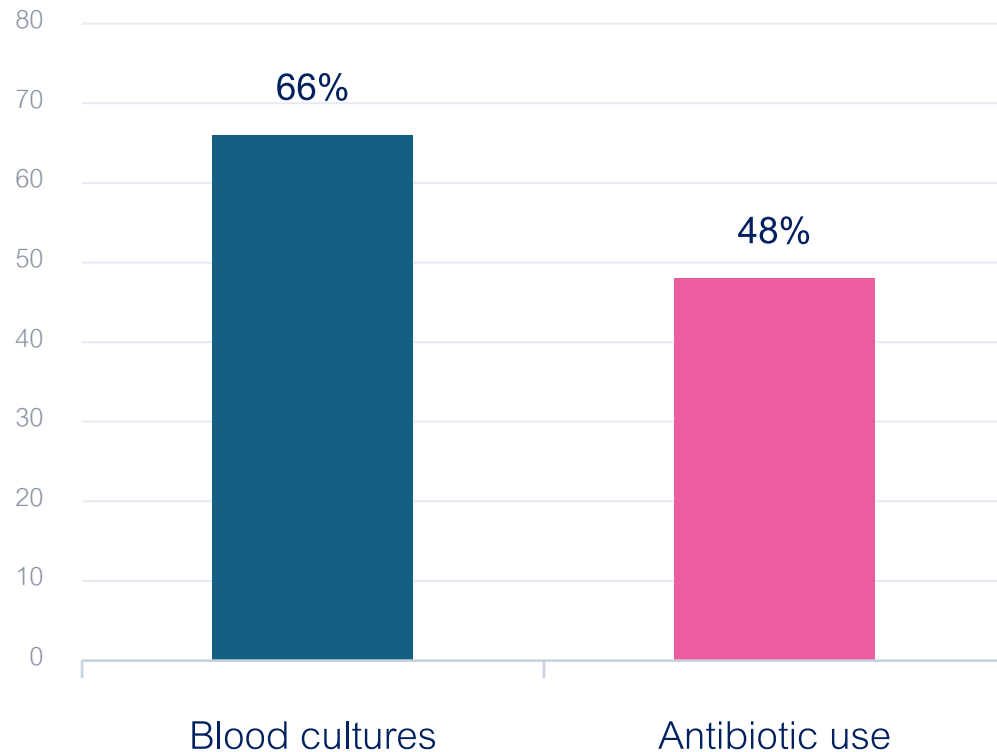
Requires: assessment at set times, standardised documentation, and clear escalation criteria



EOS risk calculator: the quantitative evidence

Compared with conventional risk assessment, with no worsening of clinical outcomes

Reduction with EOS calculator



Number needed to treat (NNT)

22

Antibiotic administration

8

Laboratory testing

7

Admission to the newborn unit

Safety

No increase in mortality, culture-confirmed infection, or readmission after discharge



Summary and application to practice

1

Assess risk quantitatively

Use the EOS calculator or systematic clinical observation instead of deciding from binary risk factors

2

Use biomarkers to stop

Two negative CRPs ≥ 24 hours apart build confidence to stop; they are not a reason to start

3

Stop within 36-48 hours

When the culture is negative and the infant is well, stopping is safe practice under AAP guidance

“Treatment driven by fear rather than evidence can cause harm that stays invisible for years”

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